

CS23633 – Cloud Computing Laboratory

Experiment 3:

Setting Up Hybrid Cloud Connectivity Using Cisco Packet Tracer (Simulation)

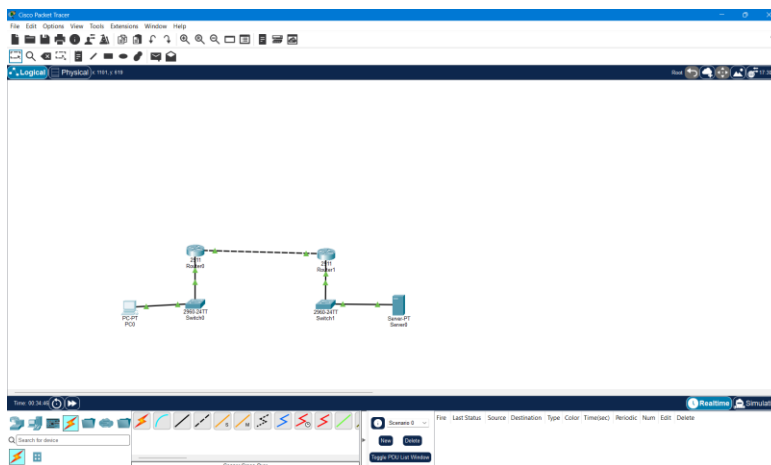
Aim:

To simulate a secure hybrid cloud connection between an on-premises data center and a cloud network using routers, switches, subnets, and routing in Cisco Packet Tracer.

Software Used:

- Cisco Packet Tracer
- Windows 11 (or your OS)

Network Topology:



Hybrid cloud network topology.

- PC
- Switch0
- Router0
- Router1
- Switch1
- Server
- All green links

Router0 Configuration

```

Router0
Physical Config CLI Attributes

IOS Command Line Interface

Router(config-if)#
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 10.10.10.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#do show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.1.1     YES manual up          up
GigabitEthernet0/1  10.10.10.1     YES manual up          down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.1.1     YES manual up          up
GigabitEthernet0/1  10.10.10.1     YES manual up          up
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 10.0.0.0 255.255.255.0 10.10.10.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#

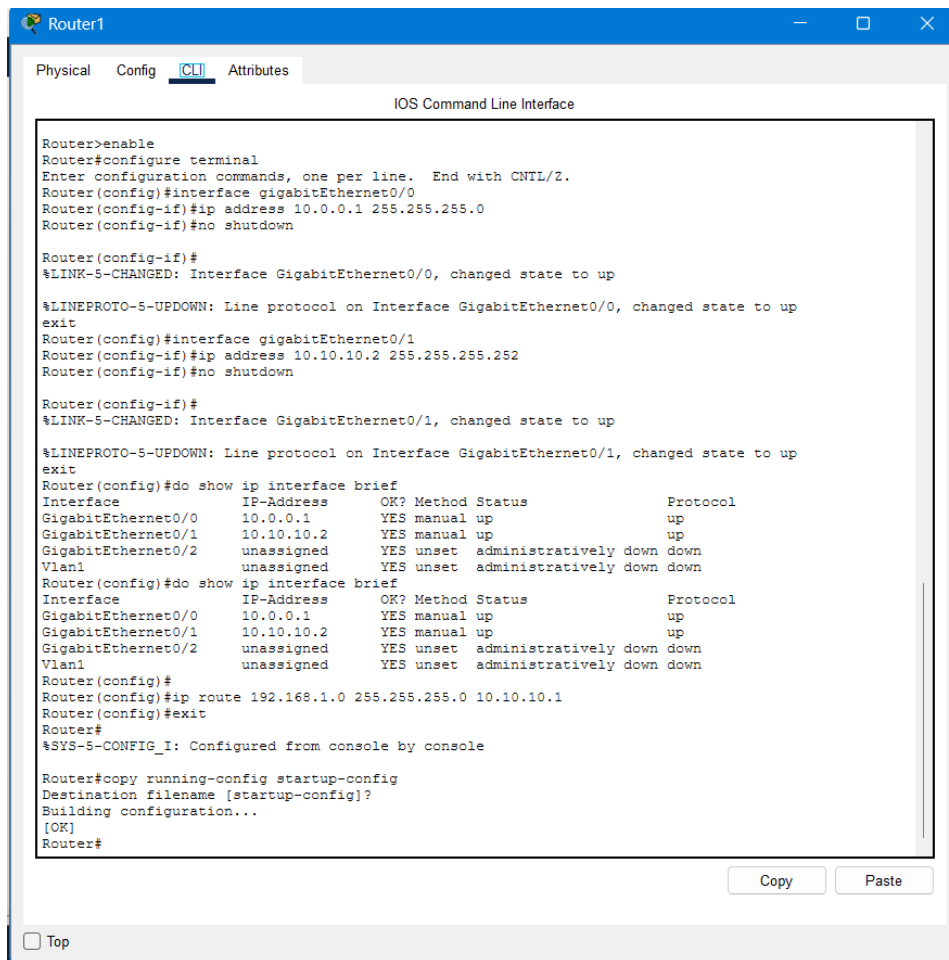
```

Copy Paste

☐ Top

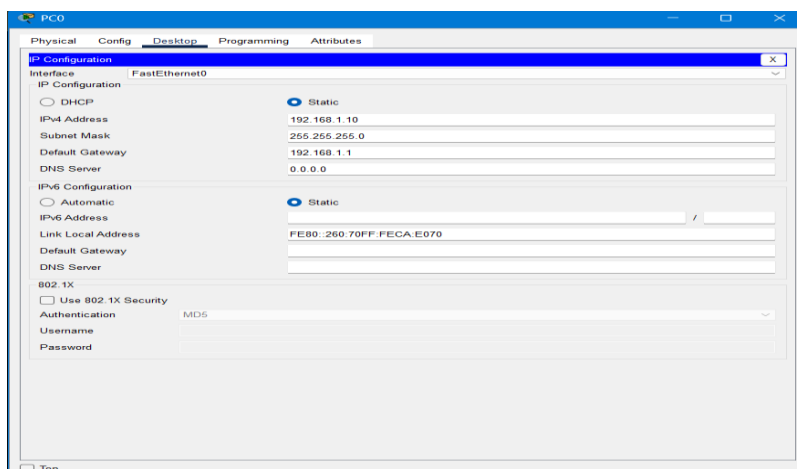
Router0 interface configuration.

Router1 Configuration:



Router1 interface configuration.

PC Configuration:



PC IP configuration.

Server Configuration:

The screenshot shows the 'Server0' configuration window with the 'Config' tab selected. The 'IP Configuration' section is active, showing options for DHCP and Static IP. The Static IP is configured with the following values:

- IPv4 Address: 10.0.0.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.0.0.1
- DNS Server: 0.0.0.0

The IPv6 Configuration section is also visible, showing options for Automatic and Static IP. The Static IP is configured with the following values:

- IPv6 Address: (empty)
- Link Local Address: FE80: 260:5CFF:FEC8:39B8
- Default Gateway: (empty)
- DNS Server: (empty)

Below the IP Configuration section, there is a checkbox for '802.1X' and a checkbox for 'Use 802.1X Security'. The 'Authentication' dropdown is set to 'MD5'. The 'Username' and 'Password' fields are empty.

Server IP configuration.

Static Route Configuration:

The screenshot shows two configuration windows for 'Router0' and 'Router1'. Both windows have the 'CLI' tab selected, showing the 'IOS Command Line Interface'.

Router0 CLI:

```
Router con0 is now available

Press RETURN to get started.

Router>
Translating "\*.domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
S    10.0.0.0/24 [1/0] via 10.10.10.2
C    10.10.10.0/30 is directly connected, GigabitEthernet0/1
L    10.10.10.1/32 is directly connected, GigabitEthernet0/1
C    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
L    192.168.1.0/24 is directly connected, GigabitEthernet0/0
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0

Router>
```

Router1 CLI:

```
Router1 con0 is now available

Press RETURN to get started.

Router1>
Translating "\*.domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

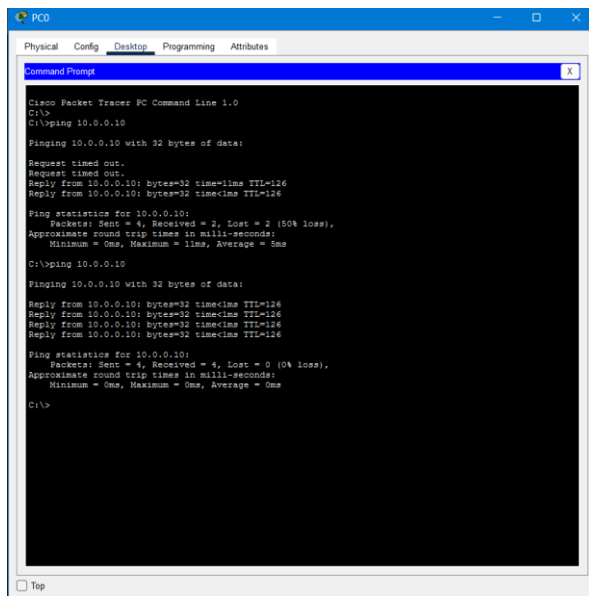
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C    10.0.0.0/24 is directly connected, GigabitEthernet0/0
L    10.0.0.1/32 is directly connected, GigabitEthernet0/0
C    10.10.10.0/30 is directly connected, GigabitEthernet0/1
L    10.10.10.2/32 is directly connected, GigabitEthernet0/1
S    192.168.1.0/24 [1/0] via 10.10.10.1

Router1>
```

Static routing configuration.

Connectivity Test:



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 5ms
C:\>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
```

Successful ping from the PC to the cloud server.

Result:

The hybrid cloud network was successfully simulated using Cisco Packet Tracer. Two routers were configured with IP addressing and static routing to interconnect the on-premises network and the cloud network. End-to-end communication between the PC and the server was verified successfully using the ping command, confirming successful hybrid cloud connectivity.